

Find the limit of the sequence if it converges; otherwise indicate divergence.

1) $a_n = (-1)^n \frac{5}{n}$

2) $a_n = \left(1 + \frac{-6}{n}\right)^n$

Find the sum of the series.

3) $\sum_{n=0}^{\infty} (-1)^n \frac{9}{4^n}$

Use partial fractions to find the sum of the series.

4) $\sum_{n=1}^{\infty} \frac{8}{n(n+3)}$

Determine if the series converges or diverges; if the series converges, find its sum.

5) $\sum_{n=1}^{\infty} \frac{3n+1}{7n-1}$

Use the integral test to determine whether the series converges.

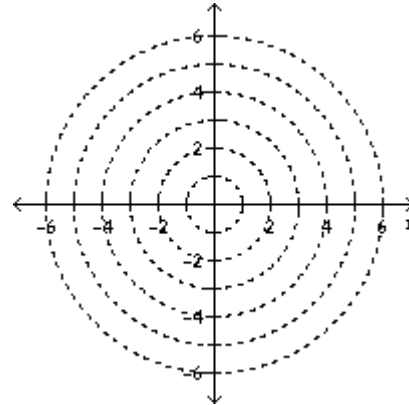
6) $\sum_{n=1}^{\infty} \frac{2n}{n^2 + 5}$

Determine convergence or divergence of the series.

7) $\sum_{n=1}^{\infty} \frac{6n^2 + 4}{n^5 + 7n + 2}$

Graph the set of points whose polar coordinates satisfy the given equation or inequality.

8) $\theta = \frac{\pi}{4}, 0 \leq r \leq 4$



Describe the graph of the polar equation.

9) $r^2 \sin 2\theta = 22$

For the given polar equation, write an equivalent rectangular equation.

10) $r^2 = 32 r \cos \theta - 6r \sin \theta - 9$

For the given rectangular equation, write an equivalent polar equation.

11) $x^2 - y^2 = 4$

Find the slope of the polar curve at the indicated point.

12) $r = 4 \cos 3\theta, \theta = \frac{\pi}{3}$

Find the area of the specified region.

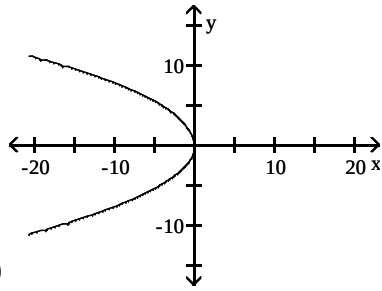
13) Inside one loop of the lemniscate $r^2 = 6 \cos 2\theta$

14) Shared by the circles $r = 8 \cos \theta$ and $r = 8 \sin \theta$

Find the length of the curve.

15) The cardioid $r = 7(1 - \sin \theta)$

Choose the equation that matches the graph.



16)

A) $x^2 = -6y$

B) $-6y^2 = x$

C) $y^2 = -6x$

D) $y^2 = 6x$

Find the focus and directrix of the parabola.

17) $x = 4y^2$

Find the standard-form equation of the ellipse centered at the origin and satisfying the given conditions.

18) An ellipse with length of major axis 16 and y-intercepts $(0, \pm 2)$

Solve the problem.

19) The ellipse $\frac{x^2}{400} + \frac{y^2}{256} = 1$ is shifted

horizontally and vertically to obtain the ellipse

$$\frac{(x+1)^2}{400} + \frac{(y+5)^2}{256} = 1.$$

Find the foci of the new ellipse.

If the equation represents a hyperbola, find the center, foci, and asymptotes. If the equation represents an ellipse, find the center, vertices, and foci. If the equation represents a circle, find the center and radius. If the equation represents a parabola, find the focus and directrix.

20) $8x^2 - y^2 - 80x + 12y - 36 = 0$

Find the eccentricity of the ellipse.

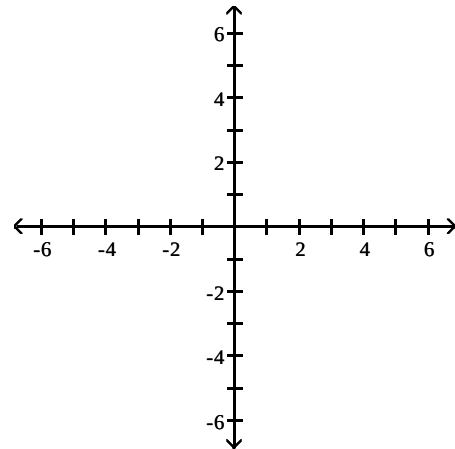
21) $16x^2 + 9y^2 = 144$

The eccentricity is given of a conic section with one focus at the origin, along with the directrix corresponding to that focus. Find a polar equation for the conic section.

22) $e = 2, y = 8$

Graph the parabola or ellipse. Include the directrix that corresponds to the focus at the origin.

23) $r = \frac{12}{4 + 4 \cos \theta}$

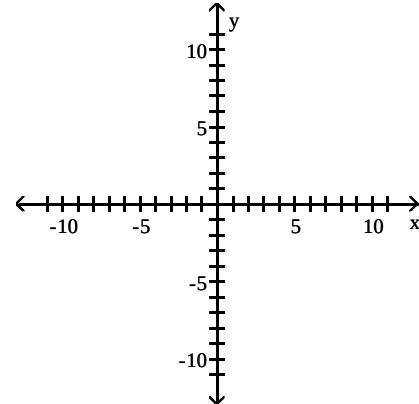


Find a Cartesian equation for the parametric equations.

24) $x = \sqrt{t}, y = 3t + 6$

Graph the pair of parametric equations in the rectangular coordinate system.

25) $x = 1 + 4 \sin t, y = 3 + 4 \cos t; 0 \leq t \leq 2\pi$



Answer Key

Testname: MA2414X4REVIEW

1) 0

2) e^{-6}

3) $\frac{36}{5}$

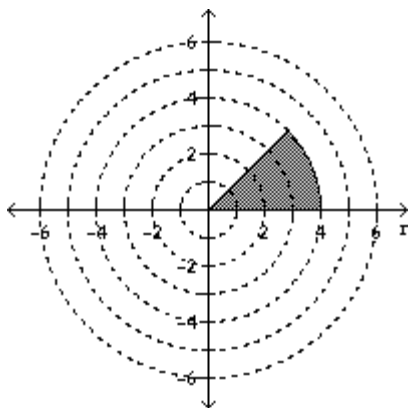
4) $\frac{44}{9}$

5) Converges; $\frac{63}{4}$

6) diverges

7) Converges

8)



9) Hyperbola with the x- and y-axes as asymptotes

10) $(x - 16)^2 + (y + 3)^2 = 256$

11) $r^2 \cos 2\theta = 4$

12) $-\frac{\sqrt{3}}{3}$

13) 3

14) $8(\pi - 2)$

15) 56

16) C

17) $(1/16, 0)$; $x = -1/16$

18) $\frac{x^2}{64} + \frac{y^2}{4} = 1$

19) $(-13, -5)$, $(11, -5)$

20) C: $(5, 6)$; F: $(5 + 15, 6)$, $(5 - 15, 6)$; A: $y - 6 = \sqrt{8}(x - 5)$, $y - 6 = -\sqrt{8}(x - 5)$

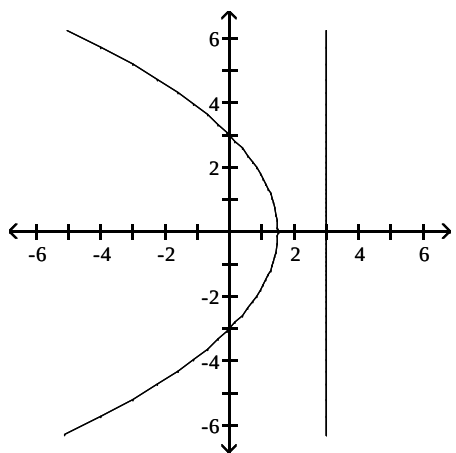
21) $\frac{\sqrt{7}}{4}$

22) $r = \frac{16}{1 + 2 \sin \theta}$

Answer Key

Testname: MA2414X4REVIEW

23)



24) $y = 3x^2 + 6$

25)

